NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2018-2019

CE1005/CZ1005 – DIGITAL LOGIC

Nov/Dec 2018 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. All questions carry equal marks.

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1. (a) Convert the following unsigned decimal numbers to hexadecimal. Show the steps clearly.

   (i) 98765

   (2 marks)

   (ii) 98.765 (your answer must have 4 significant digits)

   (3 marks)

   (b) Determine the minimum number of bits required to represent a 20-digit decimal value 3.141592… as an unsigned fixed-point binary number. Show the steps clearly.

   (5 marks)

   (c) Simplify the following Boolean expression algebraically. Give your answer in SOP (sum-of-product) form.

$$F(A,B,C,D) = \{ [(A' + C') (B' C' D)'] ' \ [(A' + B)' + C'] D' \}'$$

   (5 marks)

Note: Question No. 1 continues on Page 2

(d) Determine the signed decimal value represented by each 8-bit signed binary number below:

- 0101 0111 (sign-magnitude representation)
- 1110 0110 (sign-magnitude representation)
- 0011 0011 (2's complement representation)
- 1110 1001 (2's complement representation)

(4 marks)

(e) A logic circuit has 5 inputs A, B, C*, D*, E* where * denotes an active-low signal. Its active-low output Z* is asserted only when
- A, C*, D* are all asserted and B or E* is negated, or
- B, D*, E* are all asserted and both A, C* are negated

Using appropriate logic symbols, sketch a logic circuit diagram to show clearly how Z* is asserted. Label all signals clearly. Write a POS (product-of-sum) Boolean expression for Z*. You are not required to simplify the expression.

(6 marks)

2. (a) Perform the multiplication (-7)x(-6) as a series of signed 2's complement binary additions. Show the steps and answer clearly.

(5 marks)

(b) Figure Q2 shows (i) the desired logic waveform and (ii) the waveform produced by a push-button. Use a sketch to describe clearly how a suitable logic gate with Schmitt-trigger input may be used to obtain waveform (i) from waveform (ii).

(5 marks)

(i) Desired waveform (ii) Waveform from push-button

Figure Q2

Note: Question No. 2 continues on Page 3

(c) Table Q2 shows the voltage and current parameters of two logic devices.

Table Q2

Parameter (unit)	Device A	Device B
V_{CC} (V)	5	5
$V_{IH(min)}$ (V)	2.0	3.5
$V_{IL(max)}$ (V)	0.8	1.2
$V_{OH(min)}$ (V)	3.3	4.5
$V_{OL(max)}$ (V)	0.4	0.6
$I_{IH(max)}$ (mA)	0.6	0.7
$I_{IL(max)}$ (mA)	0.8	0.9
$I_{OH(max)}$ (mA)	17	19
$I_{OL(max)}$ (mA)	18	21

(i) Determine whether or not the devices are able to drive each other. Support your conclusion with reasons.

(4 marks)

(ii) Calculate the DC fan-out and noise margin if a device is able to drive the other.

(2 marks)

(d) Design a 4-input combinational logic circuit, F(v,a,b,c), such that it can be used in two different settings. Setting 1, selected when input v is logic 0, is described by the following canonical sum-of-minterm expression and "don't care" expression:

$F(a,b,c) = \sum m\,(0, 1, 3, 5)$
Don't_care (a,b,c) = m7

Setting 2, selected when input v is logic 1, is described by the following canonical product-of-maxterm expression and "don't care" expression:

$F(a,b,c) = \Pi\, M\,(1, 2, 3, 6)$
Don't_care (a,b,c) = M7

With the use of a clearly labelled Karnaugh map, obtain the minimum-cost SOP (sum-of-product) Boolean expression for F(v,a,b,c). Show the loops clearly.

(9 marks)

3. (a) Given the Boolean function F(a,b,c) = a'b'c' + ab'c + a'bc + abc'.

   (i) Implement F using a single 4x1 multiplexer and a single inverter.

   (6 marks)

   (ii) Determine the minimum sum-of-product expression for F.

   (5 marks)

   (iii) Write the Verilog module for F using a single assign statement.

   (6 marks)

   (b) Write the Verilog module for a 2-to-4 decoder using a ***case*** statement. Use *[1:0] in* as the input and *[3:0] out* as the output.

   (8 marks)

4. (a) Write the Verilog module for an 8-bit up/down counter (controlled by a single bit *up* input). Use a ***parameter*** statement to define the counter width (e.g. 8 bits).

   (8 marks)

   (b) A simple vending machine dispenses healthy muesli bars. The vending machine accepts only 50-cent and one-dollar coins and a muesli bar costs one dollar. If an excess amount is entered, for example 50 cents followed by one dollar, the transaction is rejected and all coins are returned.

   (i) Sketch the Moore-style state transition diagram if the FSM has 2 inputs (*fifty and dollar*), and 3 outputs (*insert_coin*, *dispense* and *reject_money*). Use states *Ready*, *Dispense*, *Reject* and any other states necessary. A reset forces the FSM to the *Ready* state with output {*insert_coin*, *dispense*, *reject_money*} = {1, 0, 0}.

   (5 marks)

   (ii) Write the Verilog module for the FSM using an active low asynchronous reset. Use a separate always block for the next state and output logic.

   (12 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.